R18

Code No: 156DX

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, August - 2022
NON-CONVENTIONAL ENERGY SOURCES
(Common to EEE, CSE, IT)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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1.a) With neat and necessary diagrams, discuss in detail about solar extraterrestrial and terrestrial solar radiation.
b) Classify the solar radiation instruments and with neat diagram discuss in detail about solar sun shine recorder. [7+8]

2.a) Define solar constant and explain its significance.
b) Discuss in detail about the environmental impacts of solar power. [7+8]

3.a) Compare flat plate and concentrating type of solar collectors.
b) Discuss in detail about Carnot cycle and its limitations. [8+7]

4.a) With neat and necessary diagram, explain the working principle of fuel cell.
b) With neat and necessary diagram, explain the working principle of MHD Engine. [8+7]

5.a) With neat and necessary diagram, explain Sensible, latent heat energy storage systems.
b) With neat and necessary diagram, explain the working principle of OTEC system. [8+7]

6.a) With neat and necessary diagram, explain the working principle of solar distillation.
b) Discuss the mini-hydel power plants, and their economics. [7+8]

7. With neat diagrams, compare the horizontal and vertical axis windmills. [15]

8.a) Discuss in detail about the process of an aerobatic digestion system.
b) Classify the geothermal resources, and brief about types of wells. [7+8]

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